

UNIVERSITY OF EASTERN AFRICA, BARATON

SCHOOL OF BUSINESS

DEPARTMENT OF INFORMATION SYSTEM AND COMPUTING

PROJECT DONE IN PARTIAL FULFILMENT OF COURSE CODE;

COSC391

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DATA MINING PROJECT PROPOSAL

PROJECT TITLE: Detection of Fraudulent Mobile Money Transactions Using Data Mining Techniques

1. **Background of the problem and problem statement**
   i. Mobile money services have become part of our everyday lives, especially in Kenya, where many people depend on services such as M-Pesa to send and receive money, pay bills and carry out business transactions. These services have made sending and receiving money easier, faster and more convenient. However, as the number of

people using mobile money continues to increase, cases of fraud have also become a concern. Some people lose money through unauthorized transactions, suspicious money transfers, account misuse and other fraudulent activities. This affects not only the customers who lose their money but also the trust people have in mobile money services.

ii. The main problem addressed is how to identify fraudulent transactions before they cause financial losses. Every day, many mobile money transactions take place, making it difficult to manually check each one and determine whether it is genuine or suspicious. Although financial service providers use security measures to detect fraud, some fraudulent transactions may still go unnoticed, especially when fraudsters use new methods. The project will use data mining to study transaction records and identify patterns that may indicate fraudulent activities. By training a machine learning model using previous transaction data, we aim to develop a system that can predict whether a transaction is likely to be fraudulent or legitimate. This can help financial service providers identify suspicious activities mor quickly and take appropriate action.

2. Methodology

The main objective of this project is to develop a machine learning model that can detect potentially fraudulent mobile money transactions. We will use the **classification technique.** This has been chosen because our aim is to group transactions into two categories: **fraudulent** and **legitimate.** The model will learn from previous transaction records where the correct results are already known. After learning from this information, it will be able to examine new transactions and predict whether they may be fraudulent or genuine. The project will involve collecting the data, understanding its contents, cleaning and preparing it, training the model and testing how well it performs.

For our dataset, we will use the publicly available PaySim Synthetic Financial Dataset for Fraud Detection, which can be obtained from Kaggle at https://www.kaggle.com/datasets/ealaxi/paysim1. The dataset contains approximately 6,362,620 transaction records and includes details such as transaction type, amount transferred, account balances before and after transactions, and whether a transaction was fraudulent. The dataset sample to be used will be about 300,000 to 500,000 to keep computation manageable. The missing information, duplicate records, incorrect entries and other errors will be checked. Information such as transaction types will be converted into a format that the machine learning algorithms can understand. The data will then be divided into training, validation and testing sets so that we can train the model and check how well it works on transactions it has not seen before.

The machine learning algorithms to be used are:

i. **Logistic Regression-** Helps to establish a basic model for identifying fraudulent transactions

ii. **Decision Tree-** Helps understand the different conditions that may indicate suspicious activity.

iii. **Random Forest-** Combines several decision trees to improve the accuracy and reliability of predictions.

The three models will be trained and compared to their results to identify which one performs best. To measure their performance, accuracy, precision, recall, F-1 score and confusion matrix will be implemented. Particular attention will be given to precision and recall because detecting fraud is more important than simply achieving a high accuracy

score. Validation process will help to select the best model before testing it using a separate set of data. The model that gives the best balance between detecting fraudulent transactions and avoiding unnecessary fraud alerts will be selected.

3. **Main Objectives of the Project**

   The main objective of this project is to develop a data mining model that can help identify fraudulent mobile money transactions. The specific objectives are:

   a. To study mobile money transaction records and identify patterns that may be linked to fraud.

   b. To clean and prepare the collected data for use in machine learning.

   c. To develop and compare three classification algorithms: Logistic Regression, Decision Tree and Random Forest.

   d. To test the models and measure how accurately they can identify fraudulent transactions.

   e. To identify the best-performing model and explain how it can help improve the detection of mobile money fraud.

4. **Expected Outcomes**

   By the end of this project, we expect to have developed a machine learning model that can identify transactions that may be fraudulent based on the information provided in the dataset. We also expect to understand some of the common patterns associated with fraudulent mobile money activities and determine which of the three algorithms performs best. The project will demonstrate how data mining can be used to solve a real-life problem by helping financial service providers identify suspicious transactions and reduce the risk of financial losses. Although our dataset is simulated, the project will give us an opportunity to understand how similar technology could be applied in real mobile money services to improve transaction security.

5. **Results and Discussion**

   After training and testing the three machine learning algorithms, we obtained different results in their ability to identify fraudulent mobile money transactions. We evaluated the model using accuracy, precision, recall and F1-score.

   - **i.** Logistic Regression achieved an accuracy of 99.91%
   - **ii.** Decision Tree achieved 99.93%
   - **iii.** Random Forest achieved the highest accuracy of 99.96%.

   In terms of precision, **Random Forest** also performed best with 94.23%, compared to 79.05% for **Logistic Regression** and 74.73% for **Decision Tree. Random Forest** achieved a recall of 75.00% and an F1-score of 83.52%, making it the strongest overall model in our comparison.

   From the results, we selected **Random Forest** as the best model for our project because it provided the best balance between identifying fraudulent transactions and avoiding unnecessary alerts on genuine transactions. The model correctly identified 147 out of 196 fraudulent transactions in the test data, while incorrectly flagging only 9 legitimate transactions as fraudulent. Although it missed some fraudulent transactions, its overall performance shows that it can be useful in identifying suspicious mobile money activities. Therefore, we can conclude that the project has achieved its main objective of developing a model that can distinguish between fraudulent and legitimate transactions. However, further improvements would be necessary before applying it to actual mobile money services.

   Model Performance Comparison

Evaluation Measure	Logistic Regression	Decision Tree	Random Forest
Accuracy	99.91%	99.93%	99.96%
Precision	79.05%	74.73%	94.23%
Recall	42.35%	69.39%	75.00%
F1-score	55.115%	71.96%	83.52%

6. **Summary and Conclusion**
   - ✓ In conclusion, this project focused on using data mining techniques to detect fraudulent mobile money transactions and help reduce financial losses. Using the PaySim dataset, we developed and compared three machine learning algorithms: **Logistic Regression, Decision Tree and Random Forest**.
   - ✓ After testing the models; Random Forest emerged as the best-performing algorithm, achieving 99.96% accuracy, 94.23% precision, 75.00% recall and an F1-score of 83.52%. These results show that Random Forest was more effective than the other two models in identifying suspicious transactions in our test data. The project has demonstrated how machine learning can be used to address the problem of mobile money fraud.
   - ✓ In the future, the model could be improved by using more diverse and real-world transaction data, testing additional algorithms and identifying new fraud patterns. We also recommend developing the model into a real-time fraud detection system that can analyse transactions as they happen and alert financial service providers whenever suspicious activities are detected.
7. **Github Repository**
   https://github.com/Kiprop-Robin/Data-Mining--Mobile-Fraud-Detection